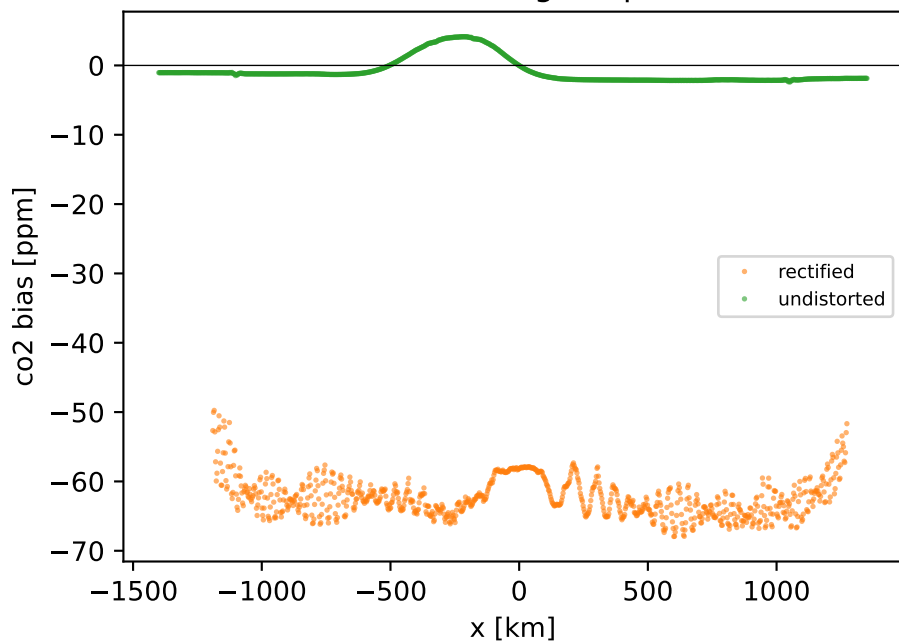
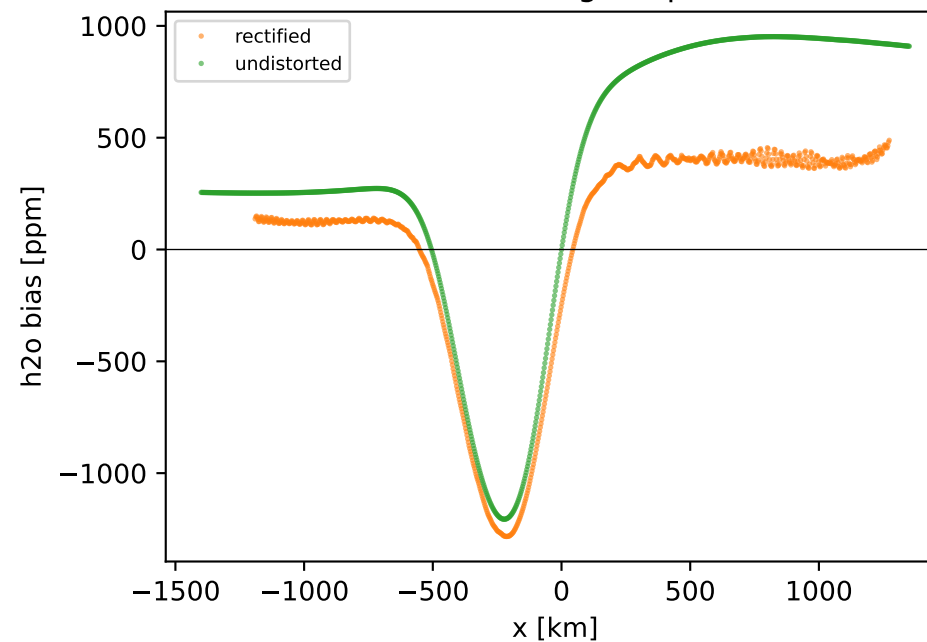


Joint FPA2 -- realistic, no noise (rectified/undistorted)

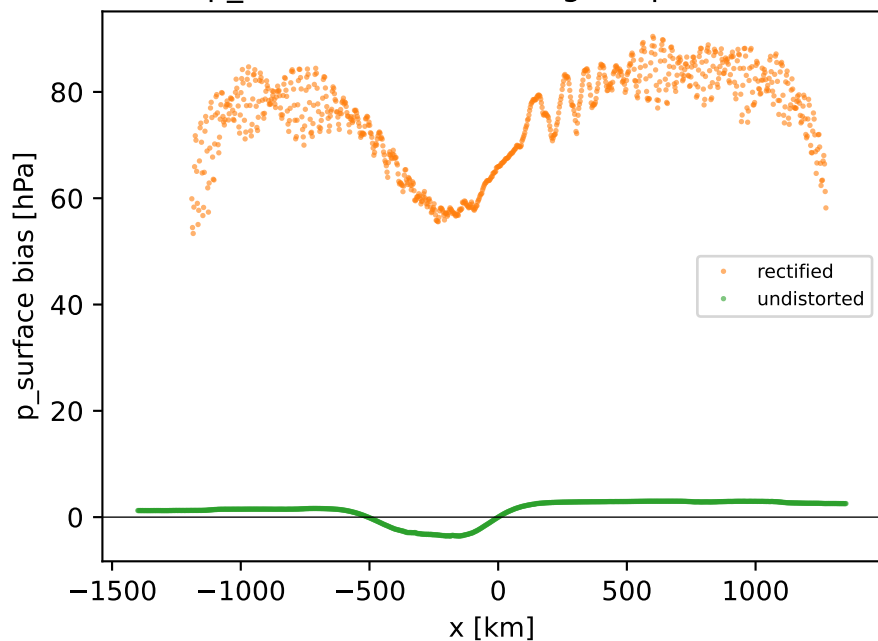
co2 bias vs along-slit position



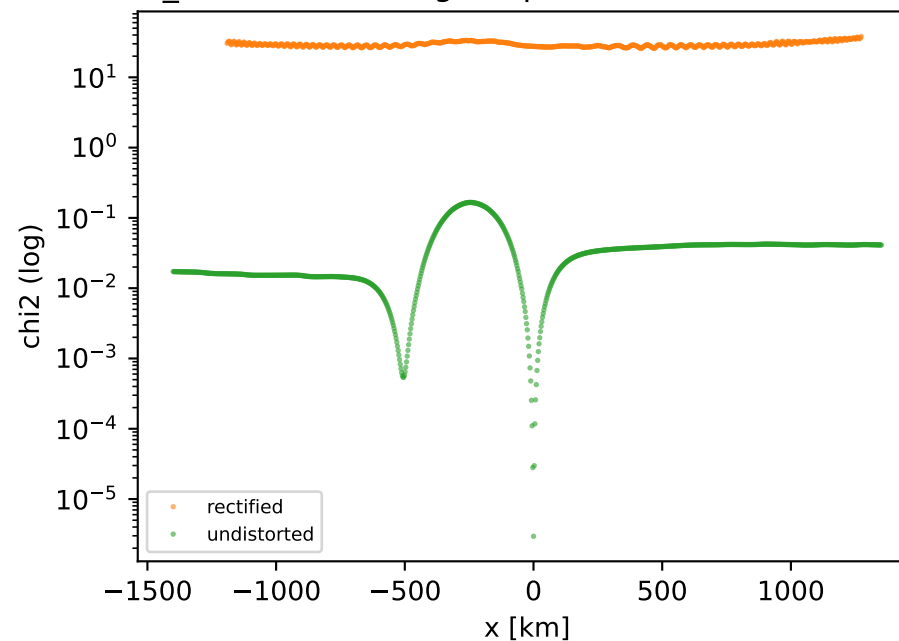
h2o bias vs along-slit position



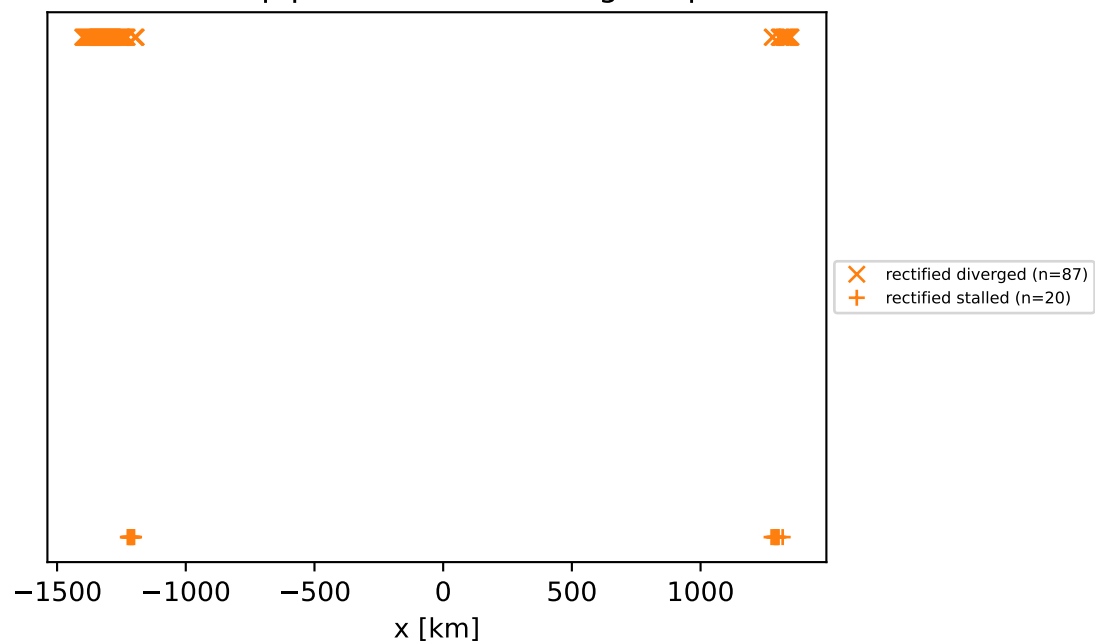
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

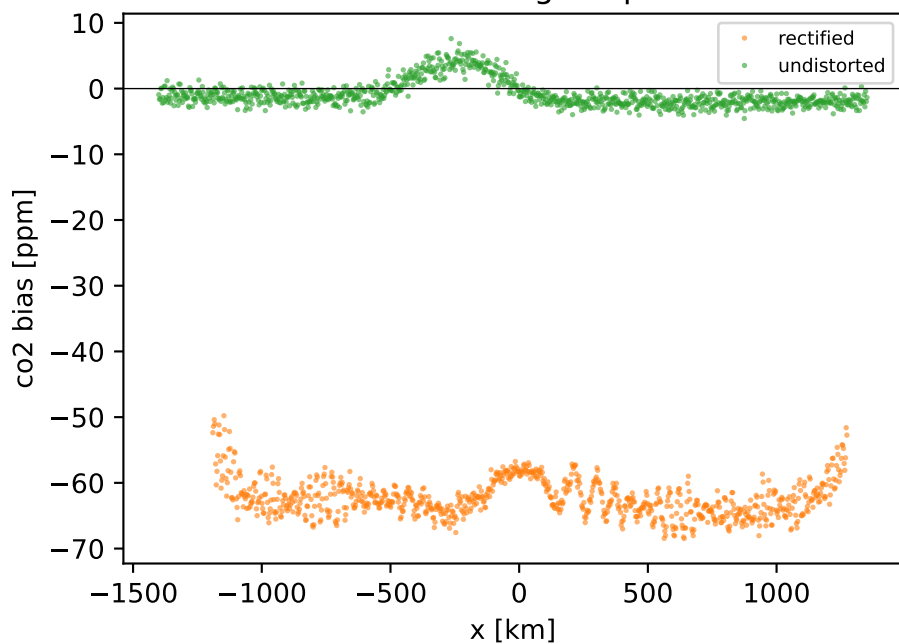


Where each pipeline fails, vs along-slit position

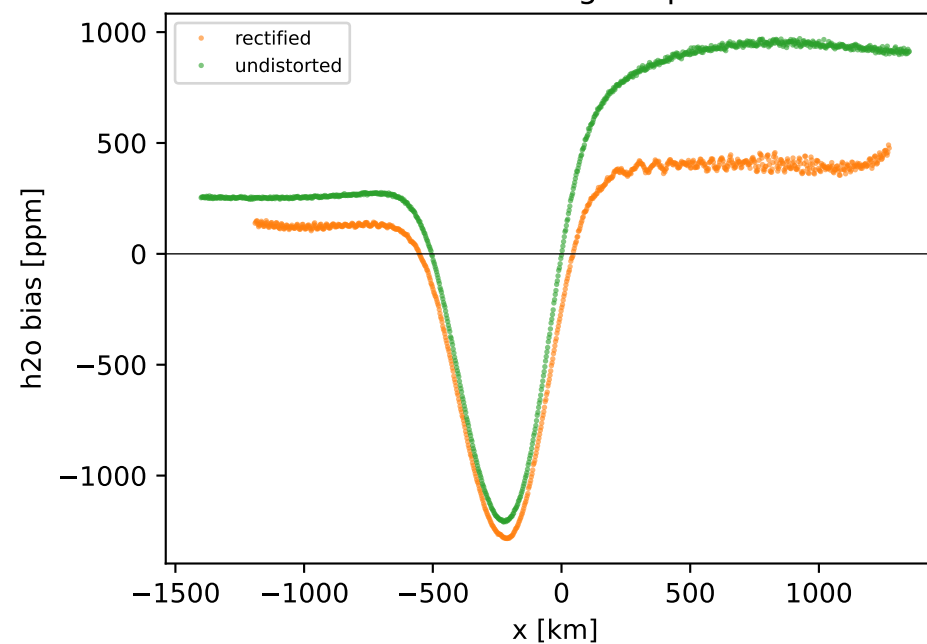


Joint FPA2 -- realistic, noise (rectified/undistorted)

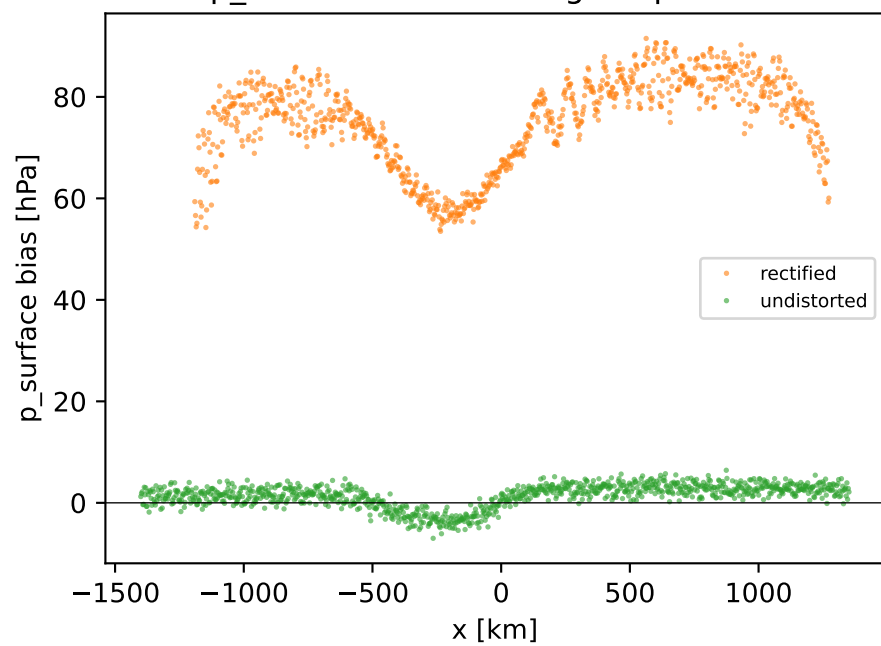
co2 bias vs along-slit position



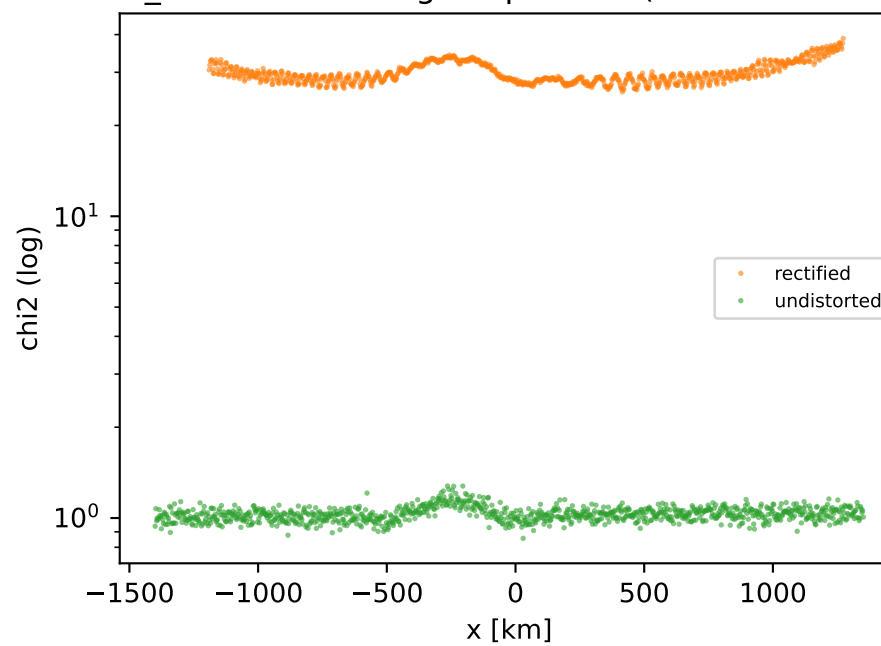
h2o bias vs along-slit position



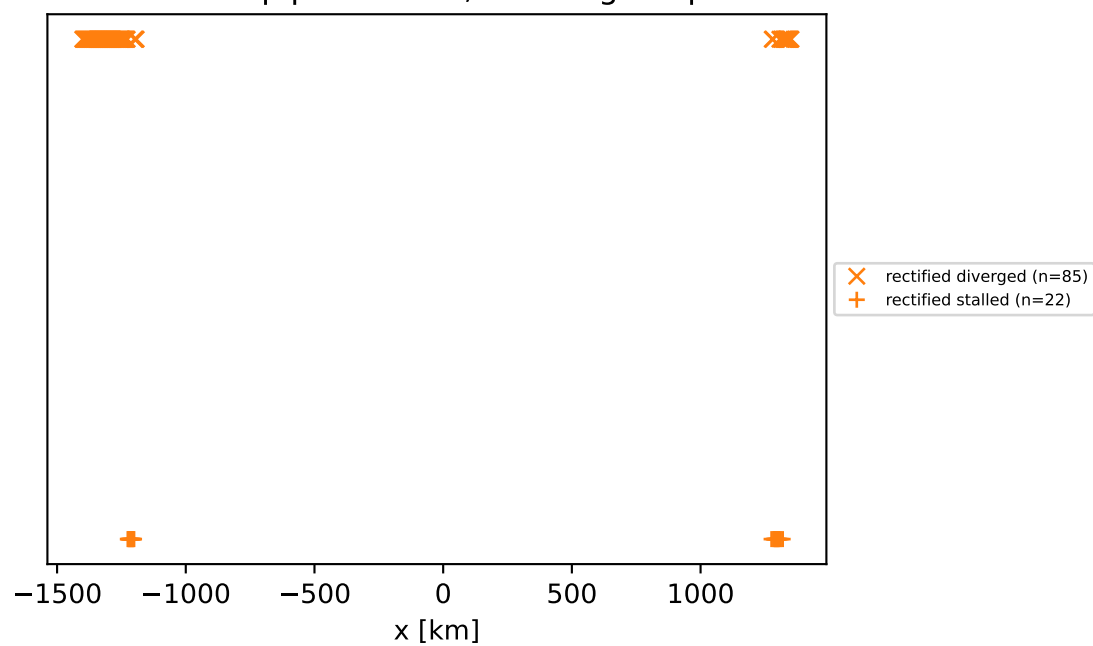
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

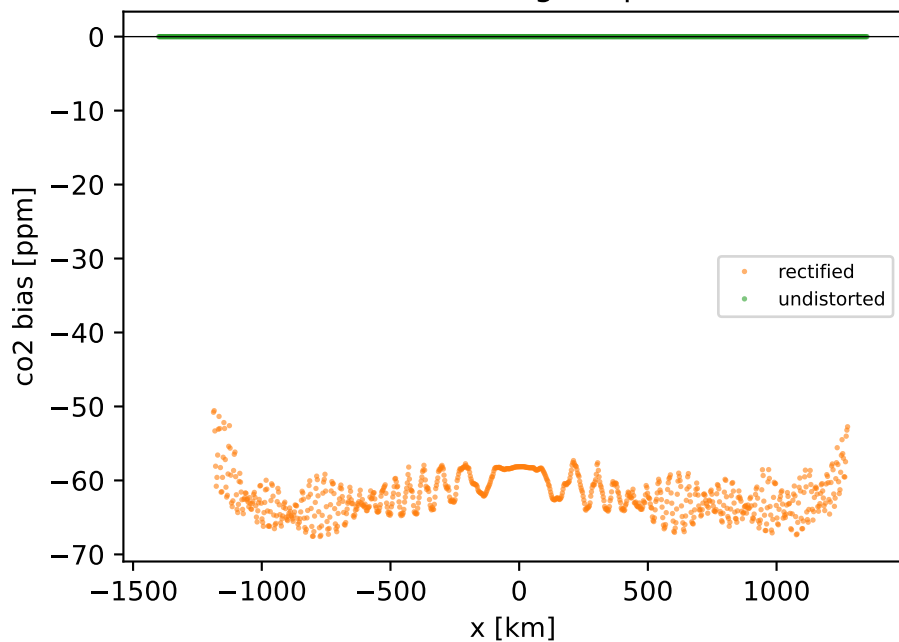


Where each pipeline fails, vs along-slit position

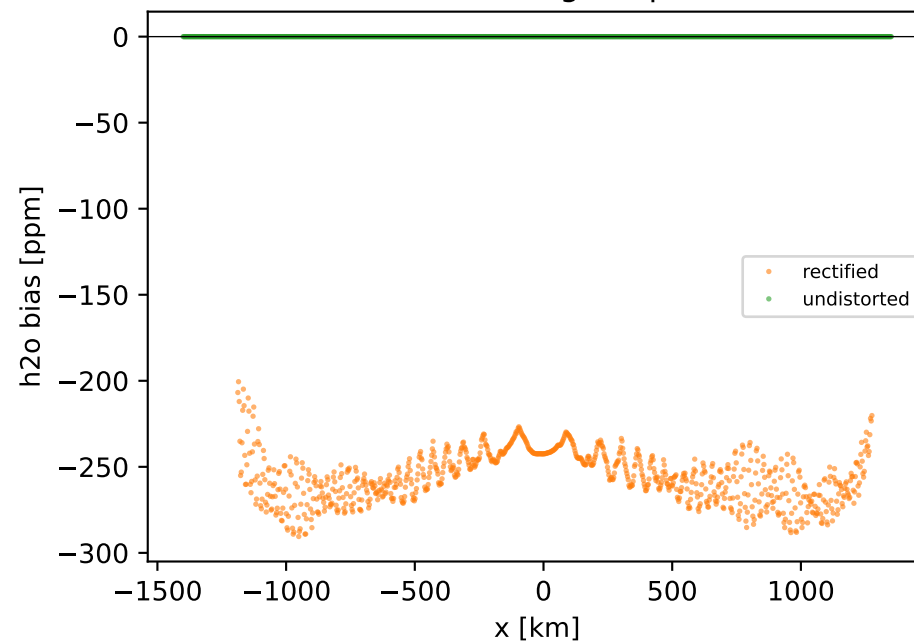


Joint FPA2 -- uniform, no noise (rectified/undistorted)

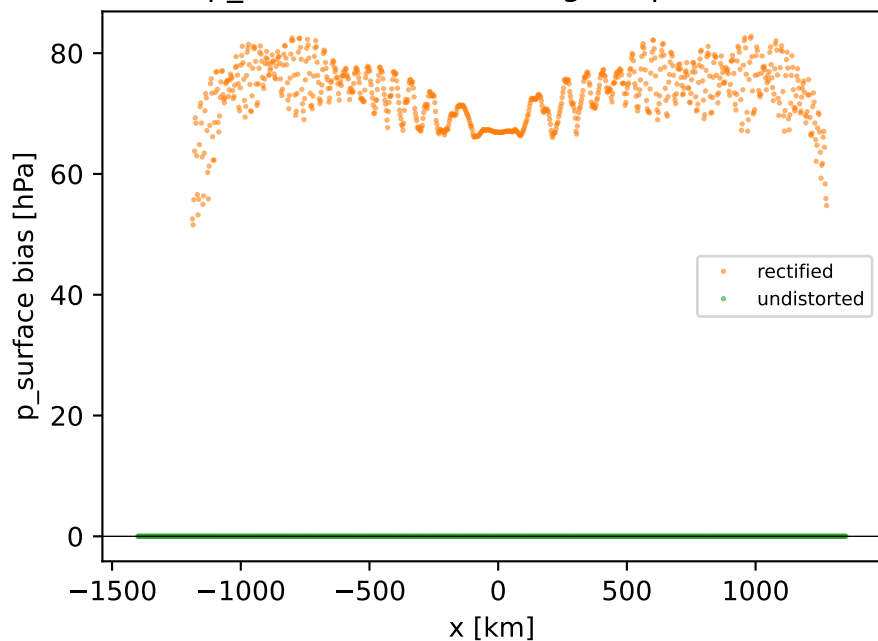
co2 bias vs along-slit position



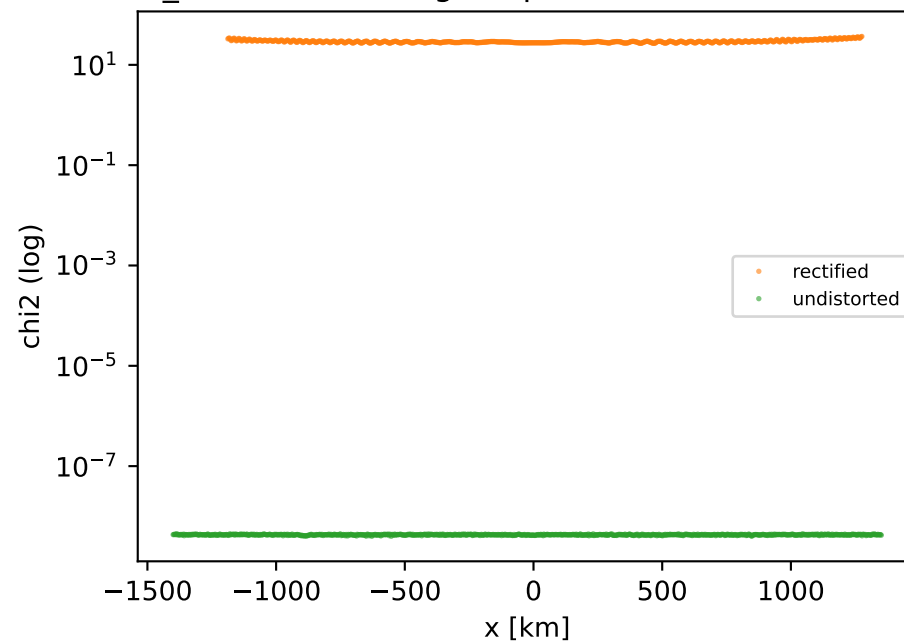
h2o bias vs along-slit position



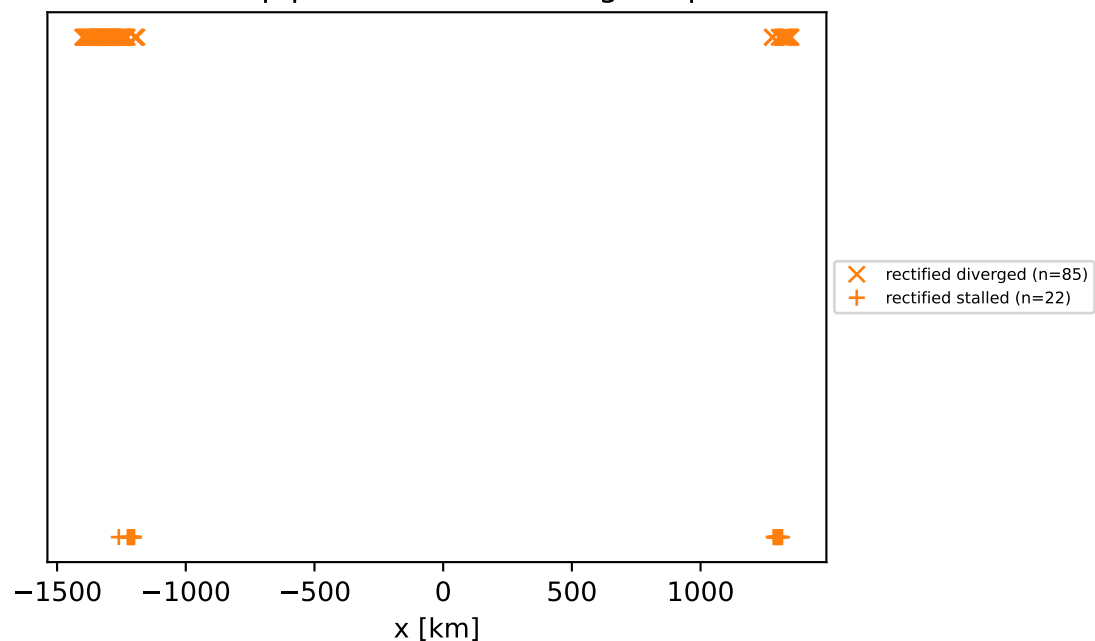
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

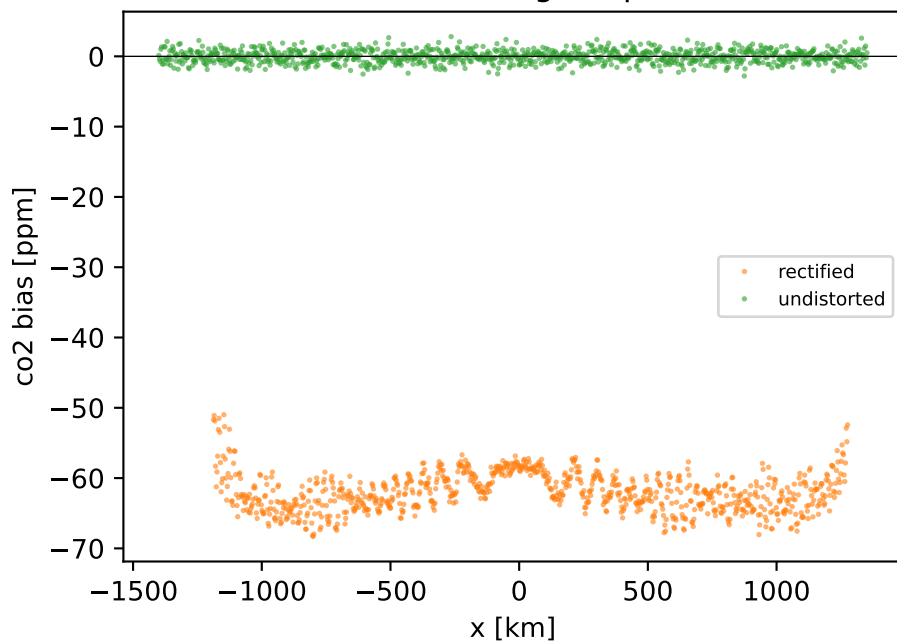


Where each pipeline fails, vs along-slit position

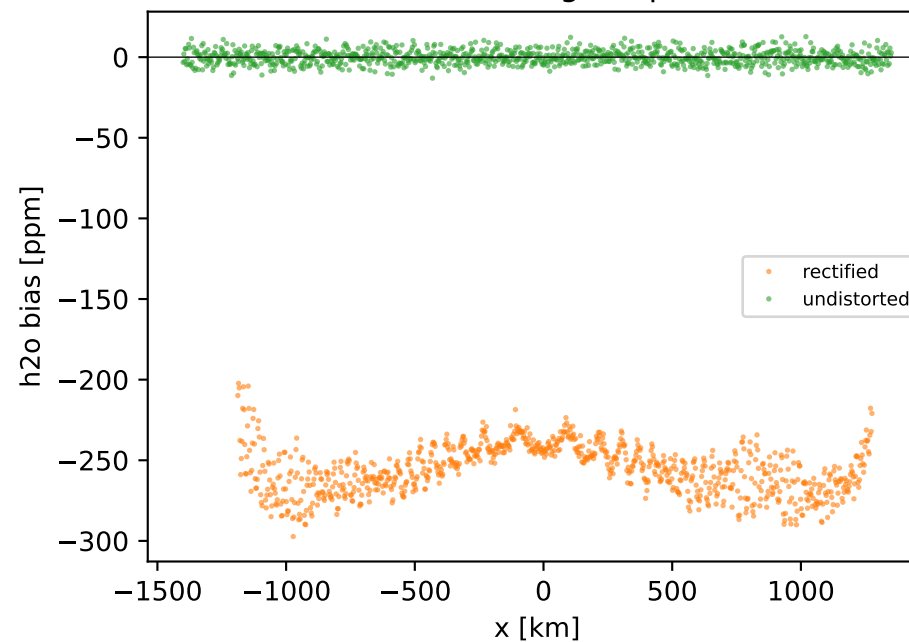


Joint FPA2 -- uniform, noise (rectified/undistorted)

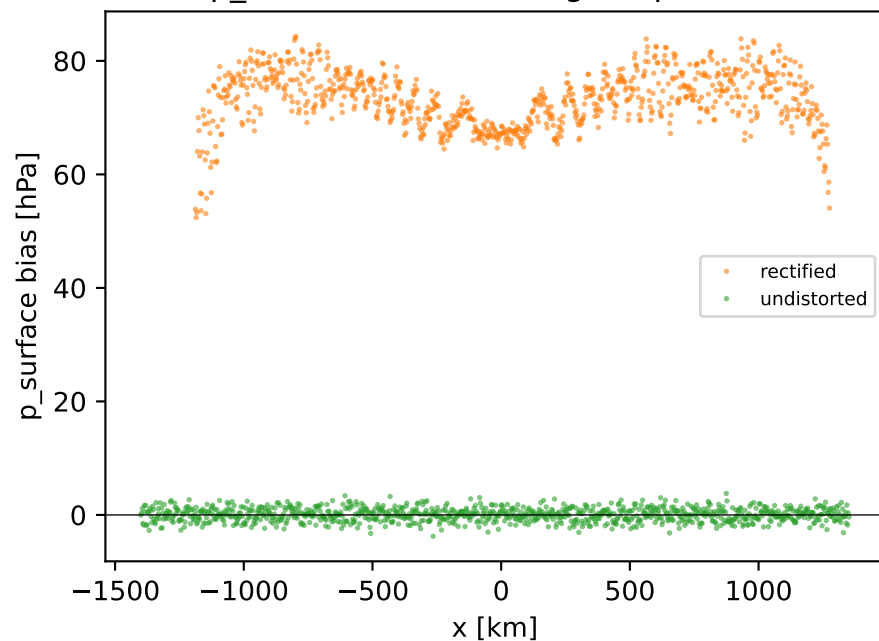
co2 bias vs along-slit position



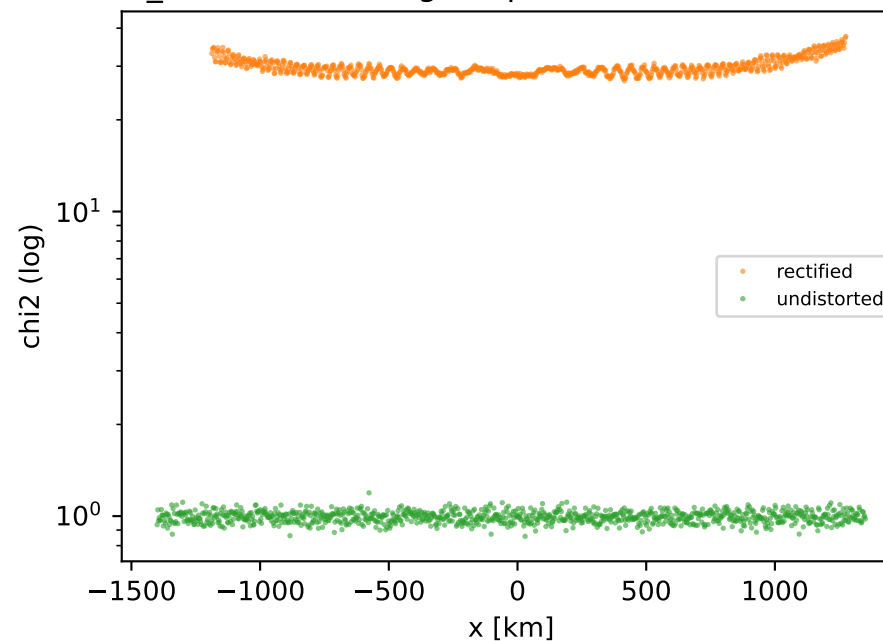
h2o bias vs along-slit position



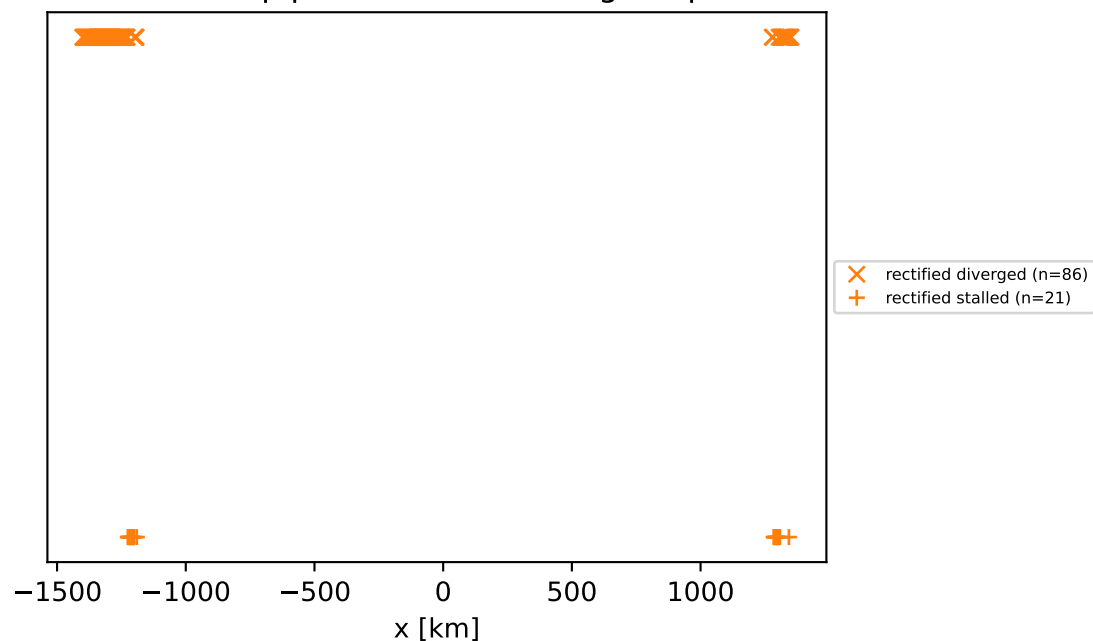
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

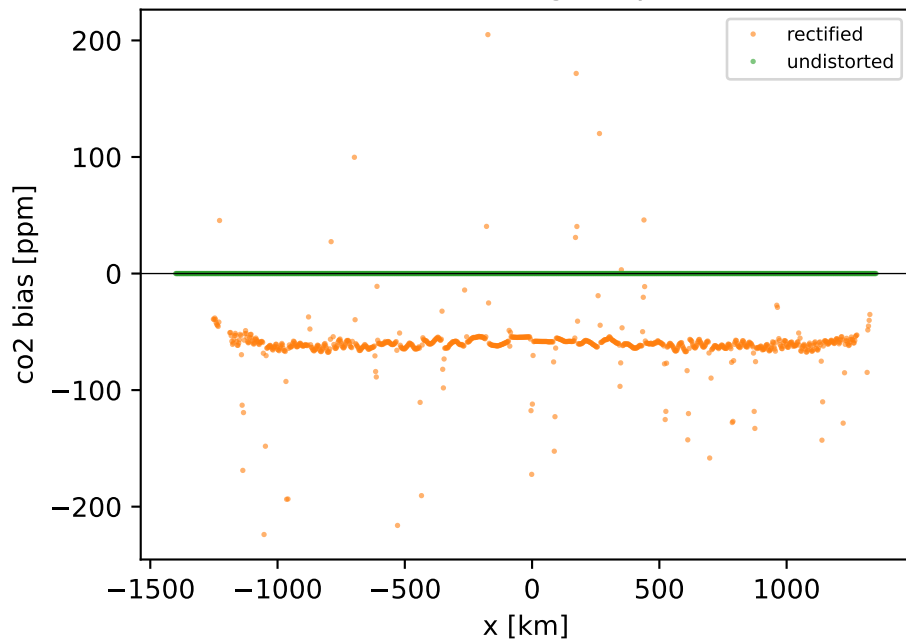


Where each pipeline fails, vs along-slit position

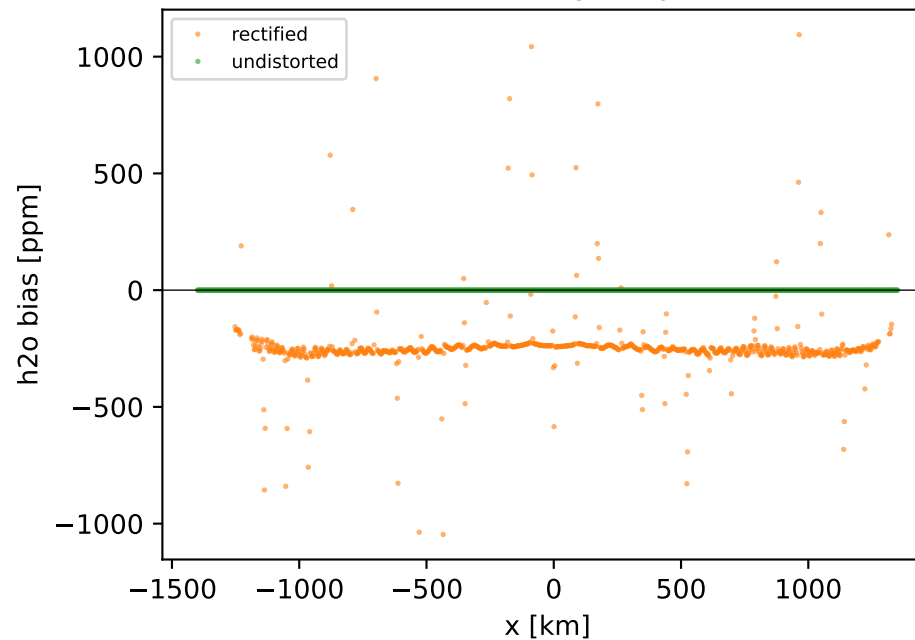


Joint FPA2 -- barcode, no noise (rectified/undistorted)

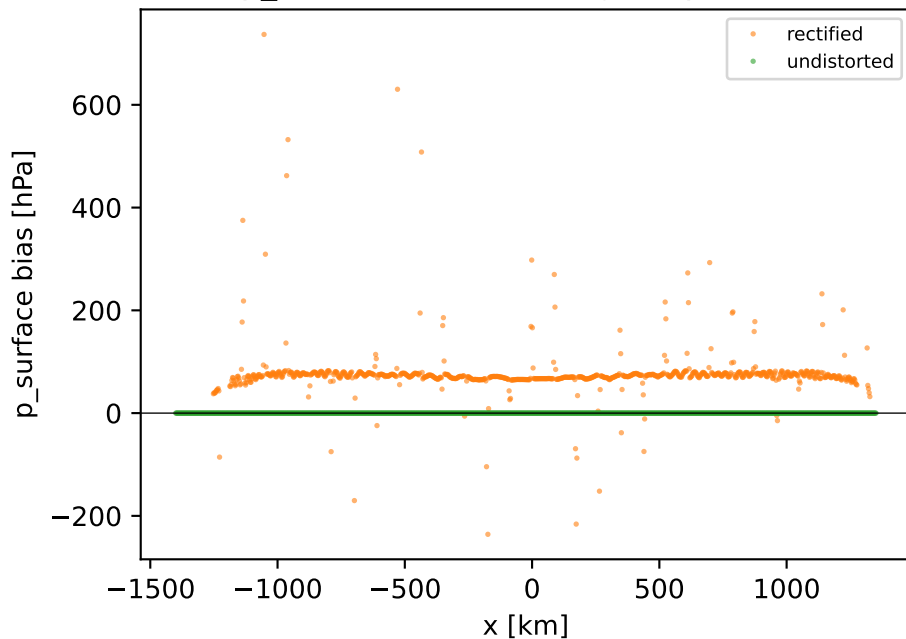
co2 bias vs along-slit position



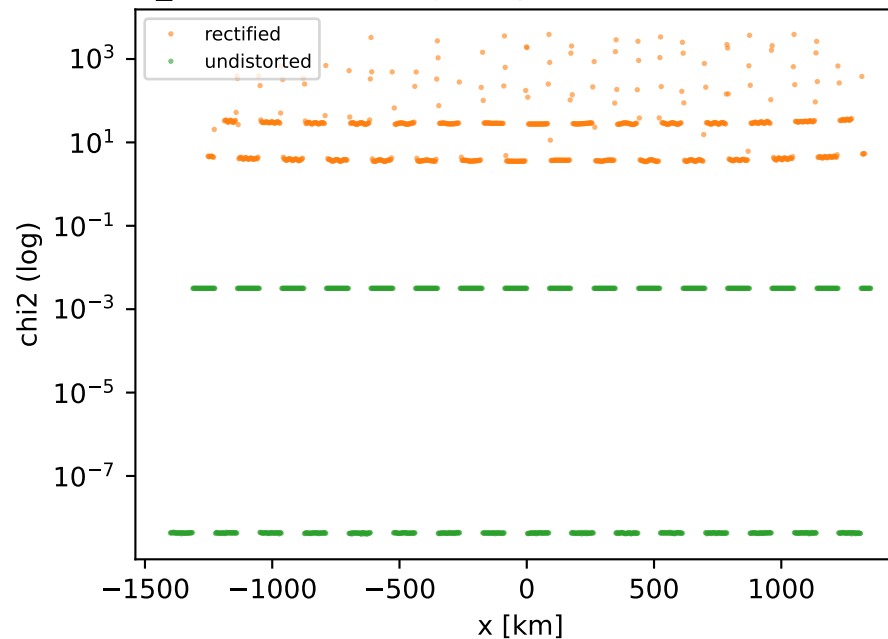
h2o bias vs along-slit position



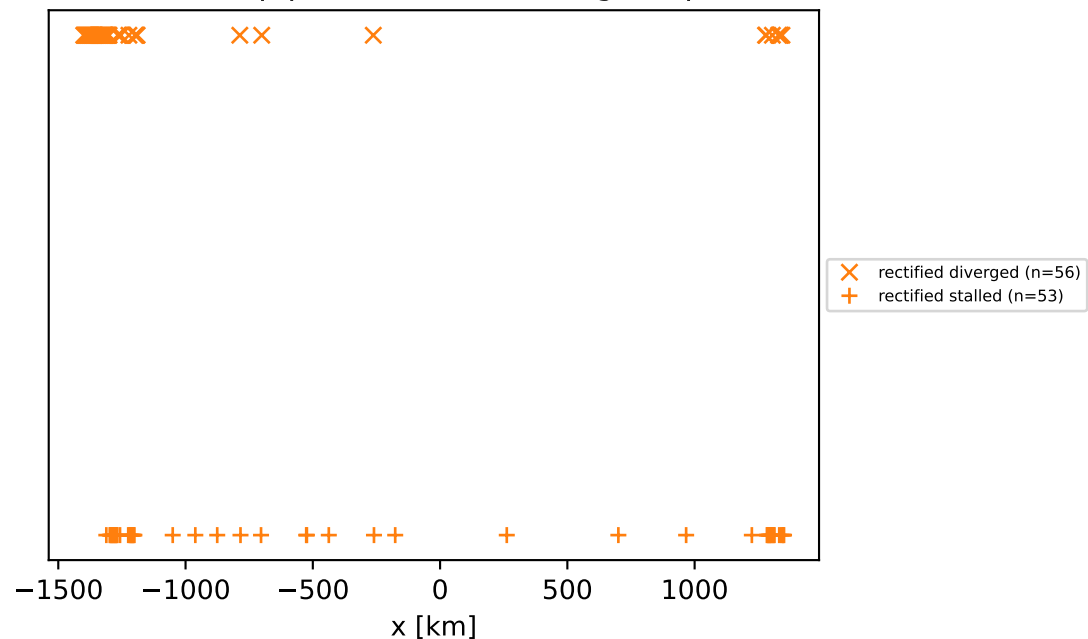
p_surface bias vs along-slit position



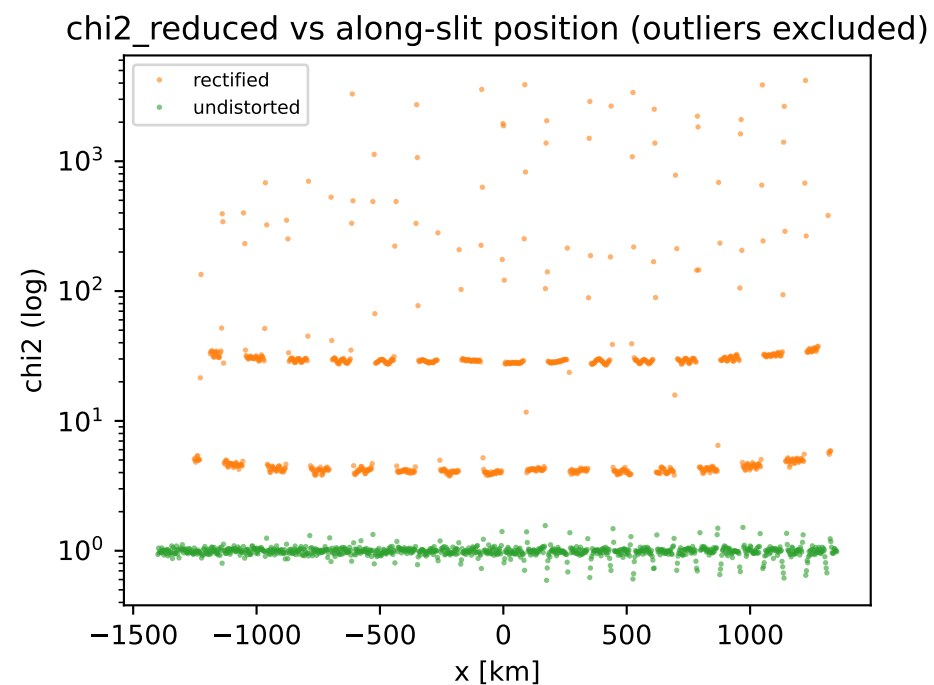
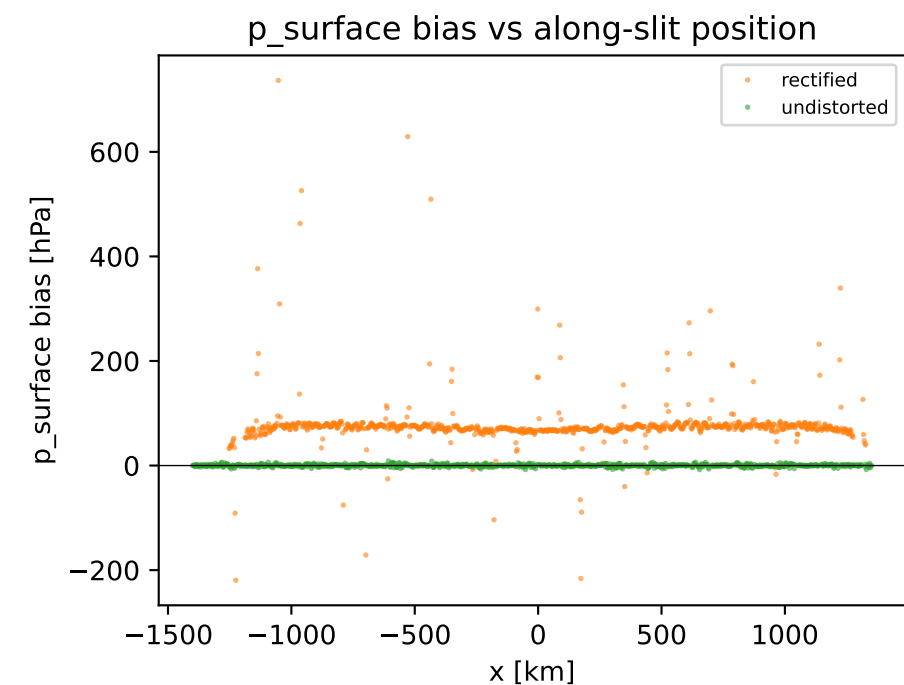
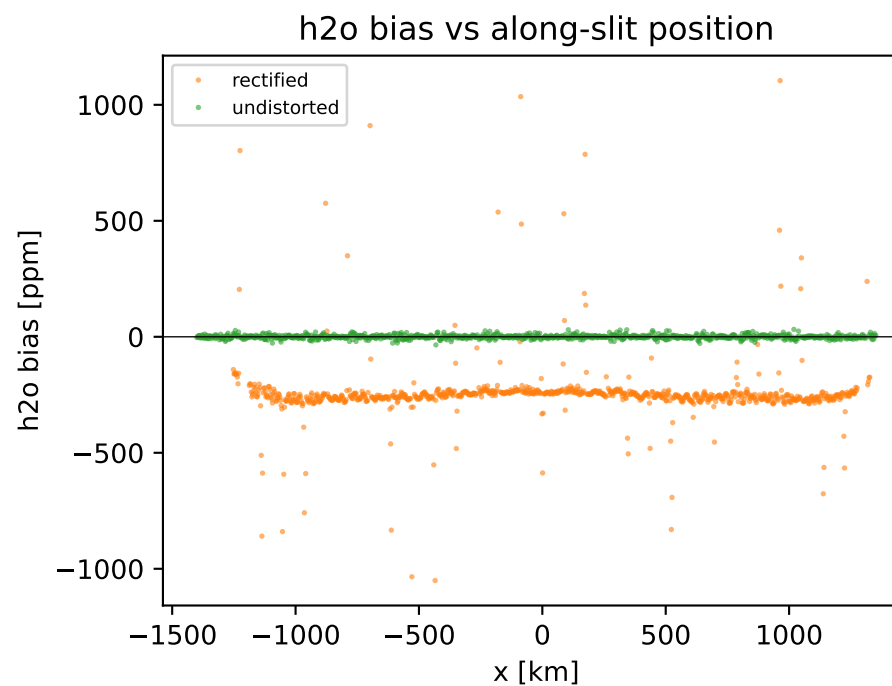
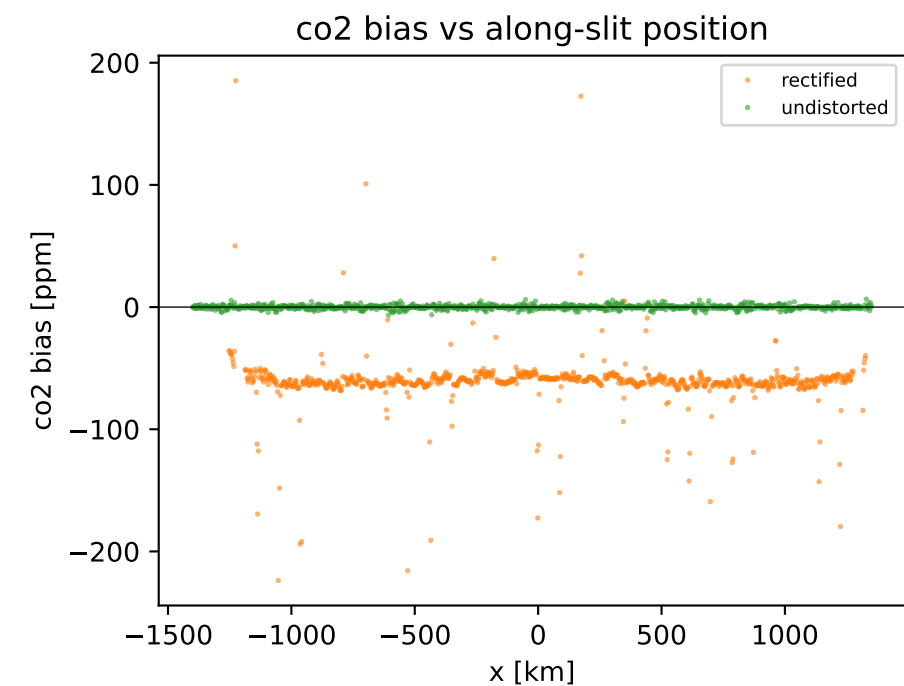
chi2_reduced vs along-slit position (outliers excluded)



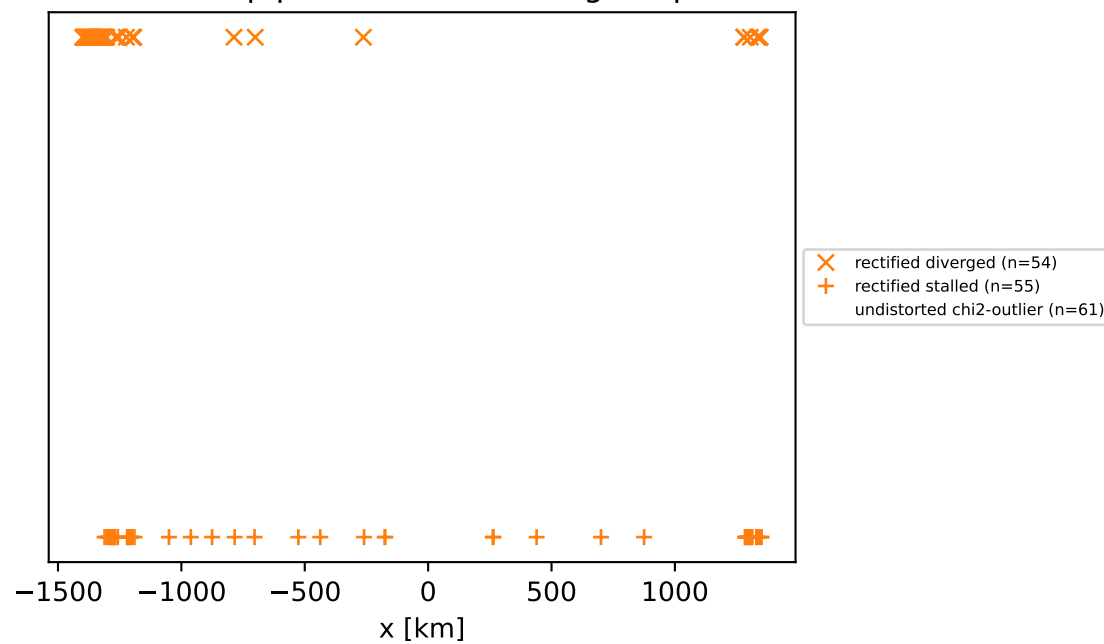
Where each pipeline fails, vs along-slit position



Joint FPA2 -- barcode, noise (rectified/undistorted)



Where each pipeline fails, vs along-slit position



FPA2 -- robust bias statistics across scenes/noise (rectified/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	rectified	917	-62.22	2.823	-62.66	2.625	0
realistic	0	h2o	rectified	917	+43.82	498.2	+138.9	373.4	0
realistic	0	p_surface	rectified	917	+75.05	8.852	+77.13	8.399	0
realistic	0	co2	undistorted	1024	-0.8539	1.79	-1.278	1.161	0
realistic	0	h2o	undistorted	1024	+362.9	618.3	+271.6	874.5	0
realistic	0	p_surface	undistorted	1024	+1.325	1.965	+1.579	1.847	0
realistic	1	co2	rectified	917	-62.26	2.866	-62.64	2.543	0
realistic	1	h2o	rectified	917	+43.71	498.2	+139.9	371.3	0
realistic	1	p_surface	rectified	917	+75.11	8.882	+76.96	8.556	0
realistic	1	co2	undistorted	1024	-0.8792	1.991	-1.422	1.28	0
realistic	1	h2o	undistorted	1024	+362.9	618.3	+271.9	875.8	0
realistic	1	p_surface	undistorted	1024	+1.37	2.284	+1.859	1.889	0
uniform	0	co2	rectified	917	-61.9	2.687	-62.14	2.635	0
uniform	0	h2o	rectified	917	-255.4	14.54	-255.8	16.13	0
uniform	0	p_surface	rectified	917	+73.07	4.864	+73.22	4.895	0
uniform	0	co2	undistorted	1024	-0.0006186	1.587e-05	-0.000617	1.582e-05	0
uniform	0	h2o	undistorted	1024	-0.002284	6.144e-05	-0.002287	5.948e-05	0
uniform	0	p_surface	undistorted	1024	+0.0005438	2.333e-05	+0.0005404	2.363e-05	0
uniform	1	co2	rectified	917	-61.94	2.742	-62.15	2.825	0
uniform	1	h2o	rectified	917	-255.5	14.99	-255.3	16.59	0
uniform	1	p_surface	rectified	917	+73.14	4.952	+73.42	5.141	0
uniform	1	co2	undistorted	1024	-0.0285	0.8973	-0.04229	0.9185	0
uniform	1	h2o	undistorted	1024	-0.08389	4.396	-0.2735	4.457	0
uniform	1	p_surface	undistorted	1024	+0.04802	1.18	+0.06559	1.222	0
barcode	0	co2	rectified	915	-60.78	23.56	-60.34	3.699	49
barcode	0	h2o	rectified	915	-244.3	137.1	-252.6	19.6	50
barcode	0	p_surface	rectified	915	+75.83	51.79	+72.25	6.153	51
barcode	0	co2	undistorted	1024	-0.0005822	4.123e-05	-0.0005852	5.262e-05	0
barcode	0	h2o	undistorted	1024	-0.002175	0.0001347	-0.002194	0.0001605	0
barcode	0	p_surface	undistorted	1024	+0.0005812	4.641e-05	+0.0005775	5.816e-05	0
barcode	1	co2	rectified	915	-61.25	22.55	-60.61	3.971	47
barcode	1	h2o	rectified	915	-245	137.8	-252.1	20.06	49
barcode	1	p_surface	rectified	915	+76.58	51.52	+72.52	6.573	49
barcode	1	co2	undistorted	963	-0.04073	1.638	-0.0882	1.217	0
barcode	1	h2o	undistorted	963	+0.0484	8.249	-0.3845	6.474	0
barcode	1	p_surface	undistorted	963	+0.08495	2.177	+0.09496	1.679	0